Supplement to "Patch-Aware KV Cache Compression for Byte-Level Language Models"

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Extra charts and the analytic compute table that did not fit in the main report.

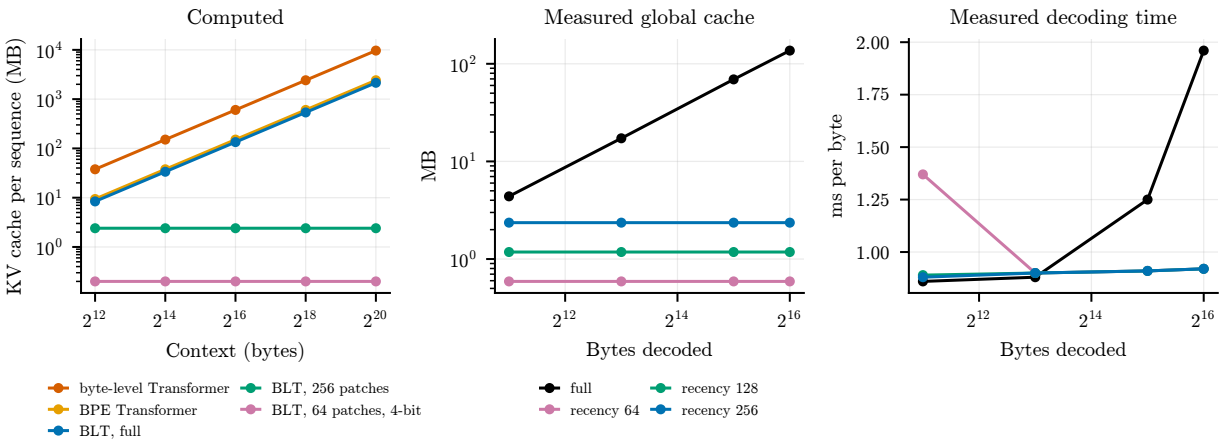


Figure S1. Cache memory. Left: computed KV size per sequence. Middle and right: measured on a GPU, one stream, real eviction of recency. A bounded cache holds the global cache at 0.59 to 2.36 MB while a full cache reaches 136.5 MB at 65,536 bytes; time per byte stays at 0.9 ms against a rise to 2.0 ms. The first recency point includes warm-up.

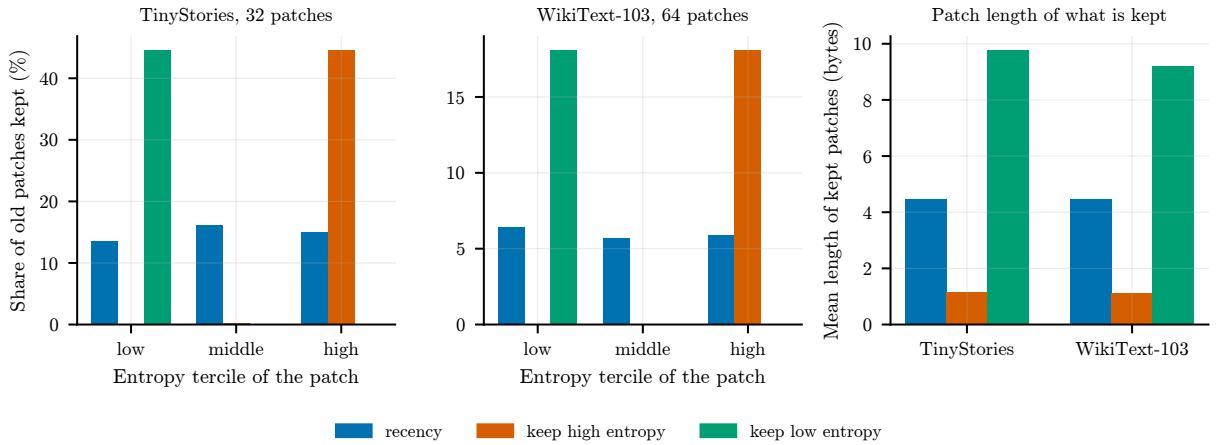


Figure S2. What each rule keeps at the close of a document. Left and middle: the share of old patches kept, by entropy tercile; keep-high-entropy keeps only high-entropy patches (and keep-low-entropy the contrary). Right: mean length of the kept patches; the entropy rule hoards one-byte fragments.

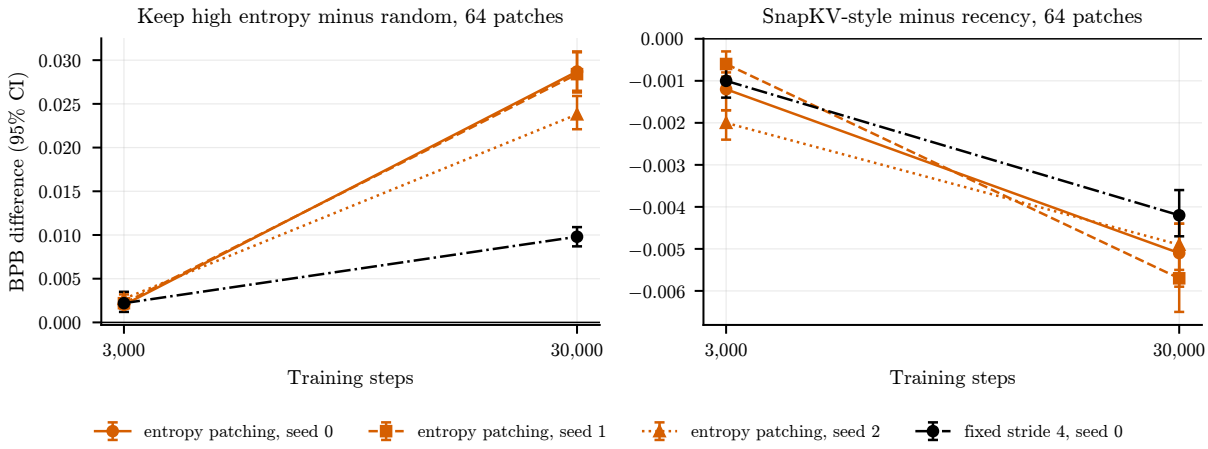


Figure S3. Ten times more training (3,000 to 30,000 steps, WikiText with code mix) makes the entropy penalty about ten times larger (left) and the SnapKV-style gain several times larger (right). Bars are 95% bootstrap intervals; three seeds of entropy patching, one of fixed stride.

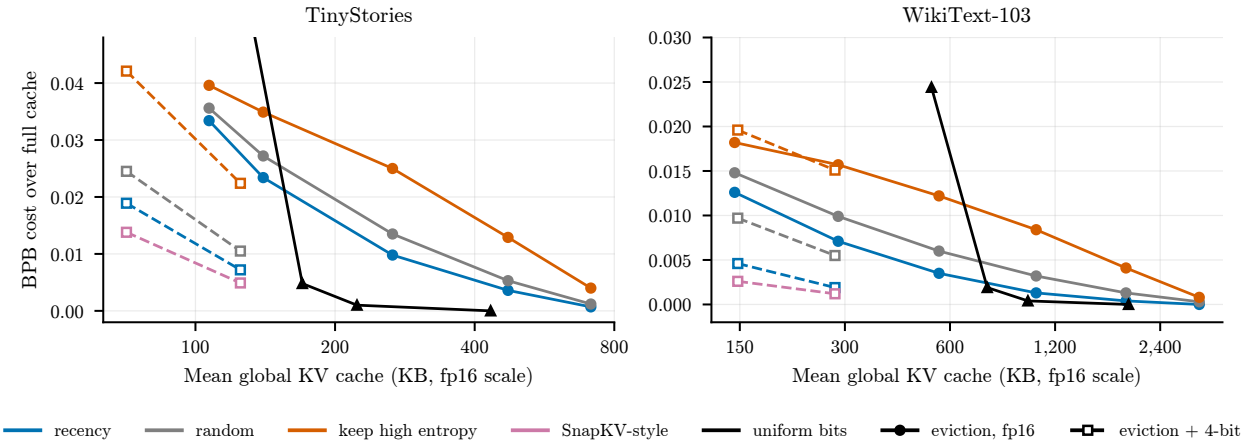


Figure S4. Cost over the full cache against cache size (eviction and uniform rows are seed 0; 4-bit rows are the three-seed mean). Keeping high entropy (orange) is the worst eviction rule at every size and never on the frontier; SnapKV-style with 4-bit (pink, hollow) is the best low-memory point. Uniform 2 bits is off scale.

Table S1. Analytic MFLOPs per generated byte and KV size (MB, fp16) at two contexts. The BPE Transformer was trained for 256 tokens (about 1,024 bytes), so its 4,096-byte rows are hypothetical.

	1,024 bytes		4,096 bytes	
Model	MFLOPs	MB	MFLOPs	MB
BLT 13.3M, full global cache	11.06	2.1	12.46	8.4
BLT, 64-patch cache	10.73	0.6	10.73	0.6
BLT, full cache, plus entropy model	19.58	2.1	20.98	8.4
BLT, 64-patch cache, plus entropy model	19.25	0.6	19.25	0.6
BPE transformer 13.8M, full cache	7.43	2.4	9.20	9.4
BPE transformer, 64-token cache	6.99	0.6	6.99	0.6
Byte-level transformer, full cache	30.87	9.4	59.18	37.7
Byte-level transformer, 64-entry cache	22.02	0.6	22.02	0.6
Byte-level transformer, 288-entry cache (same text span as 64 patches)	24.08	2.7	24.08	2.7

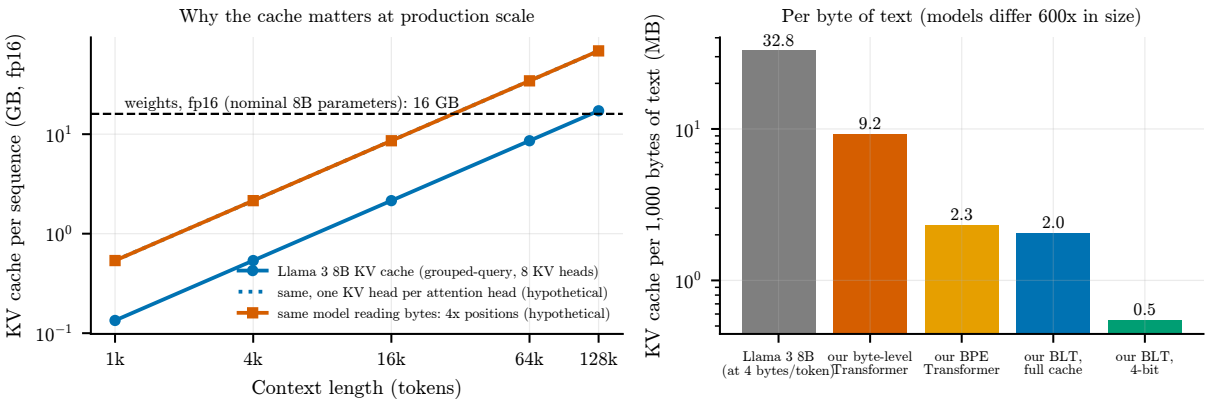


Figure S5. Left: KV cache of a production-scale model grows linearly with context and passes the size of its weights near 128k tokens (computed from the Llama 3 8B configuration: 32 layers, 8 KV heads of dimension 128; fp16). A byte-reading version of the same model would need four times as many positions (hypothetical). Right: our small models per 1,000 bytes of text; the production figure assumes 4 bytes per token and is not comparable in quality, only in cost structure. The configuration is that of the published Llama 3 8B model.

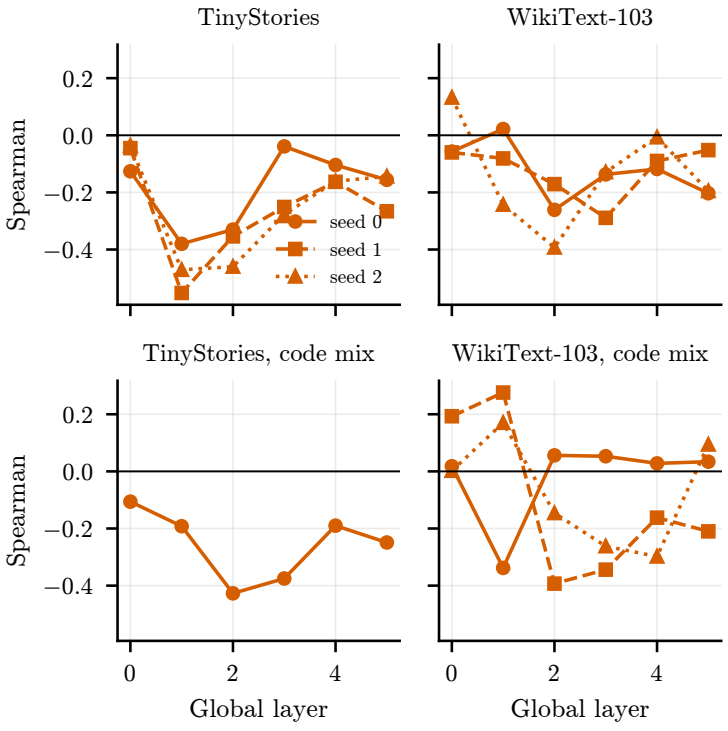


Figure S6. Spearman correlation between the entropy of an old patch (at least 8 patches back) and the attention it receives, per global layer and seed. It is negative in the middle and upper layers of most models: the model reads predictable patches more than surprising ones. Correlational evidence only; the WikiText code-mix seed 0 model is the outlier.

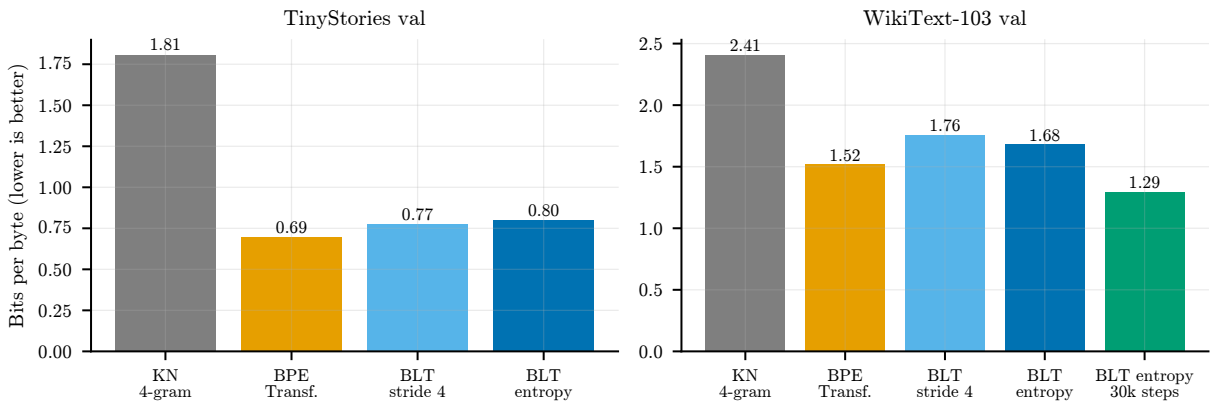


Figure S7. Bits per byte of the classical baseline (Kneser-Ney byte 4-gram), the modern baseline (BPE Transformer) and the BLT-style models. At equal training (3,000 steps) the BPE Transformer is ahead; with ten times more training the WikiText code-mix BLT is lower (a comparison that gives the BLT more training and a code mix).

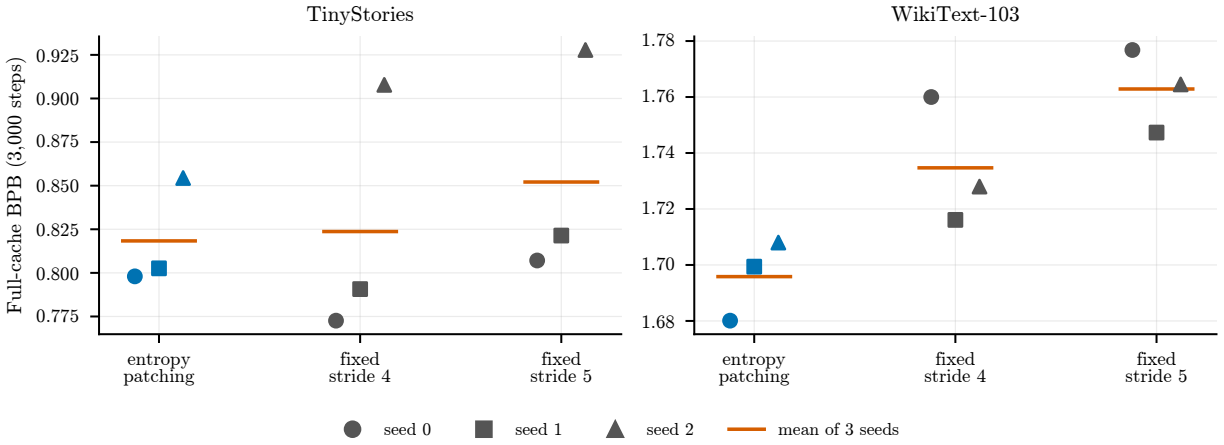


Figure S8. Why we ran three seeds. Full-cache BPB for every seed of each patching scheme: the seed spread (0.03 to 0.14) is as large as the gaps between schemes, so single-seed patching comparisons are not reliable. Entropy patching is lowest on WikiText in all three seed pairs.

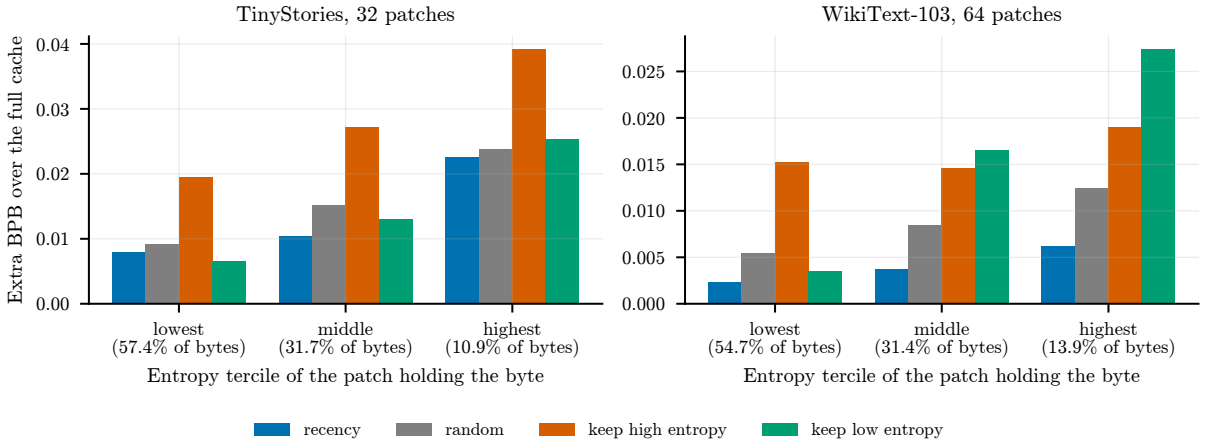


Figure S9. Extra loss over the full cache, by the entropy tercile of the patch holding the byte (first patch and 8 newest protected). Keeping high-entropy patches does not protect the high-entropy bytes and also costs the easy bytes.

Table S2. Asymmetric (KIVI-style) quantization against symmetric per-token quantization, full cache. Keys are quantized per channel in groups of 16 patches and values per token; `r8` keeps the eight newest patches in fp16. Entries are the mean over three seeds of the cost over the fp16 cache (BPB) and the mean cache size in bytes per document, scales included. At equal size the asymmetric scheme is not consistently better: on TinyStories symmetric 4-bit is both smaller and more accurate than asymmetric 2-bit.

	TinyStories		WikiText	
Scheme	bytes	ΔBPB	bytes	ΔBPB
symmetric 4-bit	222,925	+0.0012	1,003,353	+0.0004
symmetric 3-bit	170,472	+0.0064	767,270	+0.0021
symmetric 2-bit	118,019	+0.0968	531,187	+0.0300
per-token 3-bit	183,585	+0.0038	826,291	+0.0012
asymmetric 3-bit	273,102	+0.0005	1,053,976	+0.0001
asymmetric 2-bit	224,920	+0.0028	822,201	+0.0010
asymmetric 2-bit, r8	240,404	+0.0015	838,411	+0.0006

Table S3. Per-layer cache budgets with the same mean (TinyStories 32 patches, WikiText 64). The pyramid gives the lowest global layer 1.5 times the mean and the highest half of it; the inverse pyramid reverses this. Mean BPB over three seeds. The pyramid beats uniform in all 12 paired comparisons and the inverse pyramid is worse in all 12 (95% bootstrap intervals exclude zero), but the differences are small.

Dataset	Rule	uniform	pyramid	inverse
TinyStories	SnapKV-style	0.8281	0.8273	0.8336
TinyStories	recency	0.8361	0.8332	0.8421
WikiText	SnapKV-style	1.6977	1.6972	1.6993
WikiText	recency	1.7001	1.6993	1.7021